Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-2

Number of questions: **15** **Total marks: 10 X 1= 15**

1. Which of the following is a key characteristic of the Fourth Industrial Revolution?

a. Extensive use of steam engines
b. Evasion of corporate taxes
c. Widespread use of artificial intelligence
d. Introduction of assembly lines

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 1 of Week 2

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2. Which of the following is not a physical megatrend?

a. Cloud Computing
b. 3D Printer
c. Autonomous Vehicle
d. Advanced Robotics

Answer: (a)

Justification: Please refer to Slide 15 of Lecture 1 of Week 2

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3. How does Industry 4.0 enhance sustainability in manufacturing industries?

a. By increasing manual labor
b. Through globalization and emerging technologies
c. By reducing the use of digital technologies
d. Focusing on traditional manufacturing methods

Answer: (b)

Justification: Please refer to Slide 3 of Lecture 2 of Week 2

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4. What is a crucial aspect of Supply Chain Management (SCM) in sustainable manufacturing?

a. Prioritizing high-cost production
b. Ignoring environmental concerns

c. Outsourcing without strategic planning
d. Considering environmental concerns such as climate change, contamination, and resource consumption

Answer: (d)

Justification: Please refer to Slide 6 of Lecture 2 of Week 2

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5. What role does ICT play in the manufacturing industry?

a. Reducing communication within the enterprise
b. Serving as the main nervous system of any manufacturing industry
c. Discouraging the sharing of information between customer, producer, and supplier
d. Eliminating the need for enterprise resource planning

Answer: (b)

Justification: Please refer to Slide 7 of Lecture 2 of Week 2

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6. What characterizes an emerging market in the context of Industry 4.0?

a. Markets that are fully developed and static
b. Markets that meet the standards of newly developed innovative products
c. Markets that exclusively focus on traditional products
d. Markets that resist globalization

Answer: (b)

Justification: Please refer to Slide 9 of Lecture 2 of Week 2

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7. What is the primary focus of the Lean Production System?

a. Overworking the employees to boost production
b. Prioritizing customer needs
c. Increasing the price of products
d. Reducing technology use

Answer: (b)

Justification: Please refer to Slides 2 of Lecture 3 of Week 2

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8. Which among the following are aspects of socio-economic issues of sustainability in Industry 4.0?

a. Social
b. Economic
c. Environmental
d. All of the above

Answer: (d)

Justification: Please refer to Page 61 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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9. The Lean Production System mainly focuses on the customer's need. Which company is credited for the development of Lean Production System?

a. Siemens
b. Toyota Motors
c. Hitachi
d. Microsoft

Answer: (b)

Justification: Please refer to Slide 2 of Lecture 3 of Week 2

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10. What is a major concern in implementing Lean Production in the context of Industry 4.0?

a. Reducing digital technology use
b. Integrating humans in plants
c. Ignoring customer feedback
d. Avoiding environmental considerations

Answer: (b)

Justification: Please refer to Slide 9 of Lecture 3 of Week 2

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11. Which of the following is not a benefit of smart and connected products?

a. Decreased environmental impact
b. Better usage of product
c. Rapid increment in the sale price of a product
d. Smart Supply Chain

Answer: (c)

Justification: Please refer to Slide 3 of Lecture 4 of Week 2

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12. Which of the following is not a fundamental building block of a smart and connected business in the context of Industry 4.0?

    a. Blueprint of profits
    b. Customer values
    c. Key resources and processes
    d. None of the above

Answer: (d)

Justification: Please refer to Slide 5 of Lecture 4 of Week 2

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13. What is a major advantage of smart factories?

    a. Increased production costs
    b. Reduced efficiency
    c. Improvement in quality and safety
    d. Limited product variety

Answer: (c)

Justification: Please refer to Slide 4 of Lecture 5 of Week 2

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14. How is information technology utilized in smart factories?

    a. For manual data processing
    b. To avoid data analysis
    c. For smart software applications and to control processes
    d. To limit connectivity and communication

Answer: (c)

Justification: Please refer to Slide 10 of Lecture 5 of Week 2

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15. Communication, trust and capital with respect to the created values play an important role in IIoT business model.

    a. False
    b. True

Answer: (b)

Justification: Please refer to Page 109 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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